V4.8 BOM 核对清单（按位号顺序）

共 240 个位号，其中 218 个需要贴片。 贴完一个在左侧方框打勾。坐标以 PCB 左上角为原点，单位 mm。

有极性/方向 —— 贴反不工作　不贴

贴	位号	值 / 型号	封装	位置 (x, y)	阶段	备注
	C1	10uF	C_0805_2012Metric	70.8, 96.5	A 电源	
	C2	1uF	C_0603_1608Metric	85.9, 99.1	A 电源	
	C3	1uF	C_0603_1608Metric	70.8, 98.9	A 电源	
	C4	1uF	C_0603_1608Metric	78.4, 99.3	A 电源	
	C5	1uF	C_0402_1005Metric	85.9, 96.7	A 电源	
	C6	1uF	C_0603_1608Metric	92.1, 97.9	A 电源	
	C10	100nF	C_0603_1608Metric	61.0, 92.6	C 传感器 + GNSS	
	C11	100nF	C_0603_1608Metric	64.6, 87.2	C 传感器 + GNSS	
	C12	100nF	C_0603_1608Metric	56.5, 87.7	C 传感器 + GNSS	
	C13	100nF	C_0603_1608Metric	59.1, 92.6	C 传感器 + GNSS	
	C14	100nF	C_0603_1608Metric	64.6, 89.9	C 传感器 + GNSS	
	C15	4.7uF	C_0603_1608Metric	59.9, 99.3	C 传感器 + GNSS	
	C16	100nF	C_0603_1608Metric	64.6, 92.5	C 传感器 + GNSS	
	C17	100nF	C_0603_1608Metric	71.4, 81.0	C 传感器 + GNSS	
	C18	10uF	C_0805_2012Metric	74.4, 80.8	C 传感器 + GNSS	
	C19	15pF C0G	C_0603_1608Metric	88.4, 69.1	B MCU + Flash	
	C20	15pF C0G	C_0603_1608Metric	79.8, 69.1	B MCU + Flash	
	C21	100pF	C_0402_1005Metric	103.6, 59.7	E 1090 接收链	
	C22	100nF	C_0603_1608Metric	93.2, 75.8	B MCU + Flash	
	C23	100nF	C_0603_1608Metric	91.8, 81.5	B MCU + Flash	
	C24	100nF	C_0603_1608Metric	80.9, 80.2	B MCU + Flash	
	C25	100nF	C_0603_1608Metric	83.2, 71.6	B MCU + Flash	
	C26	100nF	C_0603_1608Metric	78.9, 74.6	B MCU + Flash	
	C27	100nF	C_0603_1608Metric	77.9, 82.5	B MCU + Flash	
	C28	1uF	C_0603_1608Metric	83.1, 83.3	B MCU + Flash	
	C29	1uF	C_0603_1608Metric	80.9, 83.3	B MCU + Flash	
	C30	100pF	C_0402_1005Metric	102.8, 67.5	E 1090 接收链	
	C31	100pF C0G	C_0402_1005Metric	111.3, 62.2	E 1090 接收链	
	C32	100pF C0G	C_0402_1005Metric	114.3, 62.2	E 1090 接收链	
	C33	100pF C0G	C_0402_1005Metric	120.4, 63.2	E 1090 接收链	
	C34	100pF C0G	C_0402_1005Metric	117.1, 73.4	E 1090 接收链	
	C35	100pF C0G	C_0402_1005Metric	115.1, 71.1	E 1090 接收链	
	C36	100pF	C_0402_1005Metric	106.0, 64.3	E 1090 接收链	
	C37	100nF	C_0402_1005Metric	113.9, 77.0	E 1090 接收链	
	C38	100pF	C_0402_1005Metric	111.6, 77.0	E 1090 接收链	
	C39	100pF	C_0402_1005Metric	118.2, 92.8	D 978 收发	
	C40	3.6pF	C_0402_1005Metric	110.7, 94.2	D 978 收发	
	C41	2.7pF	C_0402_1005Metric	115.9, 97.5	D 978 收发	
	C42	6.2pF	C_0402_1005Metric	115.9, 95.2	D 978 收发	
	C43	3pF	C_0402_1005Metric	115.9, 92.8	D 978 收发	
	C44	3.6pF	C_0402_1005Metric	112.8, 96.5	D 978 收发	
	C45	100pF	C_0402_1005Metric	110.7, 90.2	D 978 收发	
	C46	3pF	C_0603_1608Metric	104.9, 78.5	E 1090 接收链	
	C47	200pF	C_0603_1608Metric	104.9, 80.4	E 1090 接收链	
	C48	1uF	C_0402_1005Metric	107.4, 64.3	E 1090 接收链	
	C49	100nF	C_0603_1608Metric	76.7, 75.4	E 1090 接收链	
	C51	1nF	C_0603_1608Metric	77.0, 81.0	E 1090 接收链	

贴	位号	值 / 型号	封装	位置 (x, y)	阶段	备注
	C52	100nF	C_0402_1005Metric	99.6, 82.3	E 1090 接收链	
	C53	100pF	C_0402_1005Metric	93.6, 62.9	E 1090 接收链	J7测IFA要求后级断路；U17等未贴时C53可保留
	C54	100pF	C_0402_1005Metric	103.9, 61.7	E 1090 接收链	
	C55	100pF	C_0603_1608Metric	97.6, 66.5	E 1090 接收链	
	C56	100pF	C_0603_1608Metric	95.0, 66.5	E 1090 接收链	
	C57	100pF	C_0402_1005Metric	68.3, 81.0	C 传感器 + GNSS	
	C58	100pF	C_0402_1005Metric	57.7, 76.5	C 传感器 + GNSS	
	C59	100pF	C_0402_1005Metric	57.6, 69.8	C 传感器 + GNSS	
	C60	22uF	C_0805_2012Metric	99.3, 98.0	D 978 收发	
	C61	100nF	C_0603_1608Metric	102.7, 99.1	D 978 收发	
	C62	100nF	C_0603_1608Metric	105.4, 98.3	D 978 收发	
	C63	100nF	C_0603_1608Metric	107.1, 84.6	D 978 收发	
	C64	100nF	C_0603_1608Metric	100.2, 85.8	D 978 收发	
	C65	100nF	C_0603_1608Metric	99.0, 94.7	D 978 收发	
	C66	22uF	C_0805_2012Metric	102.9, 97.2	D 978 收发	
	C67	1uF	C_0603_1608Metric	103.6, 85.8	D 978 收发	
	C68	18pF	C_0603_1608Metric	113.2, 85.0	D 978 收发	
	C69	18pF	C_0603_1608Metric	110.1, 85.0	D 978 收发	
	C70	47nF	C_0603_1608Metric	135.7, 80.2	A 电源	
	C71	4.7uF(带电源版)	C_0603_1608Metric	133.4, 79.0	A 电源	
	C72	10uF	C_1206_3216Metric	130.9, 79.2	A 电源	
	C73	10uF	C_0805_2012Metric	141.3, 88.1	A 电源	
	C74	10uF	C_0805_2012Metric	143.3, 84.9	A 电源	
	C75	10uF	C_0805_2012Metric	139.1, 88.1	A 电源	
	C76	22uF	C_1206_3216Metric	135.2, 96.8	A 电源	
	C77	22uF 16V X5R	C_1206_3216Metric	141.2, 99.7	A 电源	
	C78	1uF	C_0603_1608Metric	134.3, 93.8	A 电源	
	C79	1uF	C_0603_1608Metric	130.6, 65.0	A 电源	
	C80	1uF 25V	C_0603_1608Metric	136.3, 74.6	A 电源	
	C81	470pF C0G	C_0402_1005Metric	115.2, 64.5	E 1090 接收链	
	C82	100nF	C_0603_1608Metric	91.4, 67.5	B MCU + Flash	
	C83	100nF	C_0603_1608Metric	91.4, 69.8	B MCU + Flash	
	C84	1uF	C_0603_1608Metric	83.3, 85.6	B MCU + Flash	
	C85	100nF	C_0603_1608Metric	113.7, 80.5	B MCU + Flash	
	C86	100nF	C_0603_1608Metric	121.4, 88.1	B MCU + Flash	
	C87	10nF C0G	C_0603_1608Metric	84.0, 88.5	B MCU + Flash	
	C88	100pF	C_0603_1608Metric	126.4, 90.7	G 航空波段 AIRBAND	
	C89	18pF	C_0603_1608Metric	128.7, 93.9	G 航空波段 AIRBAND	
	C91	10uF	C_0805_2012Metric	129.0, 108.2	G 航空波段 AIRBAND	
	C92	10uF	C_0805_2012Metric	121.3, 100.8	G 航空波段 AIRBAND	
	C93	10uF	C_0805_2012Metric	124.2, 109.4	G 航空波段 AIRBAND	
	C94	100nF	C_0603_1608Metric	120.5, 109.4	G 航空波段 AIRBAND	
	C95	100nF	C_0603_1608Metric	117.2, 99.6	G 航空波段 AIRBAND	
	C96	100nF	C_0603_1608Metric	128.0, 102.1	G 航空波段 AIRBAND	
	C97	1uF	C_0603_1608Metric	129.8, 102.1	G 航空波段 AIRBAND	
	C98	100nF	C_0603_1608Metric	131.6, 102.1	G 航空波段 AIRBAND	
	C99	100nF	C_0402_1005Metric	133.6, 100.1	G 航空波段 AIRBAND	
	C100	100nF	C_0603_1608Metric	119.5, 100.7	G 航空波段 AIRBAND	
	C101	1uF	C_0603_1608Metric	81.1, 90.7	A 电源	
	C102	1uF	C_0603_1608Metric	85.4, 95.1	A 电源	

贴	位号	值 / 型号	封装	位置 (x, y)	阶段	备注
	R1	10k	R_0603_1608Metric	55.2, 84.5	C 传感器 + GNSS	
	R2	4.7k	R_0603_1608Metric	64.1, 97.8	C 传感器 + GNSS	
	R3	4.7k	R_0603_1608Metric	64.1, 95.2	C 传感器 + GNSS	
	R4	1k	R_0603_1608Metric	82.0, 69.1	B MCU + Flash	
	R5	10k	R_0603_1608Metric	80.0, 73.2	B MCU + Flash	
	R6	10k	R_0603_1608Metric	87.9, 86.4	B MCU + Flash	
	R7	5.1k DNP(带电源版)	R_0603_1608Metric	141.4, 62.5	不贴（DNP）	
	R8	5.1k DNP(带电源版)	R_0603_1608Metric	139.4, 62.5	不贴（DNP）	
	R9	27R	R_0603_1608Metric	87.9, 83.3	B MCU + Flash	
	R10	27R	R_0603_1608Metric	85.3, 83.3	B MCU + Flash	
	R11	3.32k	R_0402_1005Metric	105.1, 59.7	E 1090 接收链	
	R17	10k	R_0603_1608Metric	122.0, 97.4	D 978 收发	
	R18	10k	R_0603_1608Metric	91.4, 71.4	E 1090 接收链	
	R19	52.3R	R_0402_1005Metric	115.1, 73.4	E 1090 接收链	
	R21	0R	R_0603_1608Metric	105.0, 72.6	E 1090 接收链	
	R22	0R	R_0603_1608Metric	100.3, 66.5	E 1090 接收链	
	R23	0R	R_0603_1608Metric	93.2, 77.2	E 1090 接收链	
	R24	0R DNP(旁路外接)	R_0402_1005Metric	101.1, 63.1	不贴（DNP）	
	R25	0R DNP(旁路板载)	R_0402_1005Metric	95.5, 62.9	不贴（DNP）	
	R26	10k	R_0603_1608Metric	67.3, 73.7	C 传感器 + GNSS	
	R27	10k	R_0603_1608Metric	67.3, 71.1	C 传感器 + GNSS	
	R30	1k DNP	R_0402_1005Metric	96.8, 77.2	不贴（DNP）	
	R31	1k	R_0603_1608Metric	101.6, 75.7	E 1090 接收链	
	R32	10k	R_0603_1608Metric	104.9, 75.7	E 1090 接收链	
	R33	100k	R_0603_1608Metric	100.9, 82.7	E 1090 接收链	
	R34	10k	R_0603_1608Metric	77.0, 73.2	E 1090 接收链	
	R35	10k	R_0603_1608Metric	77.0, 78.4	E 1090 接收链	
	R36	1k DNP	R_0603_1608Metric	96.5, 81.8	不贴（DNP）	
	R37	0R	R_0603_1608Metric	137.0, 66.4	A 电源	
	R38	180R	R_0603_1608Metric	133.1, 88.2	A 电源	
	R39	5.62k	R_0603_1608Metric	136.2, 88.2	A 电源	
	R40	31.6k	R_0603_1608Metric	134.6, 88.2	A 电源	
	R41	470k	R_0603_1608Metric	139.1, 98.6	A 电源	
	R42	150k	R_0603_1608Metric	137.4, 98.6	A 电源	
	R43	1k	R_0603_1608Metric	127.5, 65.0	A 电源	
	R44	10k	R_0603_1608Metric	137.0, 69.5	A 电源	
	R45	10k	R_0603_1608Metric	129.2, 84.0	A 电源	
	R46	10k	R_0603_1608Metric	129.2, 86.5	A 电源	
	R47	10k	R_0603_1608Metric	98.5, 91.7	D 978 收发	
	R48	0R	R_0603_1608Metric	126.8, 67.0	A 电源	
	R49	0R	R_0603_1608Metric	126.8, 69.4	A 电源	
	R50	30k	R_0603_1608Metric	86.5, 92.8	B MCU + Flash	
	R51	10k	R_0603_1608Metric	85.5, 88.5	B MCU + Flash	
	R52	10k	R_0402_1005Metric	64.4, 85.0	C 传感器 + GNSS	
	R53	10k	R_0402_1005Metric	64.3, 83.0	C 传感器 + GNSS	
	R54	10R	R_0402_1005Metric	116.5, 77.0	E 1090 接收链	
	R55	10k	R_0603_1608Metric	57.4, 89.8	C 传感器 + GNSS	
	R56	10k	R_0603_1608Metric	98.5, 90.0	D 978 收发	
	R57	33R	R_0603_1608Metric	104.0, 82.5	E 1090 接收链	

贴	位号	值 / 型号	封装	位置 (x, y)	阶段	备注
	R58	10k	R_0603_1608Metric	132.5, 108.2	G 航空波段 AIRBAND	
	R59	10k	R_0603_1608Metric	132.5, 105.6	G 航空波段 AIRBAND	
	R60	0R	R_0603_1608Metric	124.0, 101.5	G 航空波段 AIRBAND	
	R61	10k	R_0402_1005Metric	131.1, 97.9	G 航空波段 AIRBAND	
	R63	2.2k	R_0603_1608Metric	78.4, 97.8	A 电源	
	R64	1k	R_0603_1608Metric	145.8, 80.3	A 电源	
	R65	1k	R_0603_1608Metric	145.8, 84.2	A 电源	
	L1	18nH 0402CS-18NXGRW	L_0402_1005Metric	110.4, 59.8	E 1090 接收链	
	L2	33nH	L_0402_1005Metric	57.7, 75.4	C 传感器 + GNSS	
	L7	6.8uH	L_0805_2012Metric	97.3, 94.7	D 978 收发	
	L8	100nH	L_0402_1005Metric	119.3, 90.3	D 978 收发	
	L9	7.5nH	L_0402_1005Metric	112.3, 94.8	D 978 收发	
	L10	27nH	L_0402_1005Metric	110.9, 95.7	D 978 收发	
	L11	6.8nH	L_0402_1005Metric	118.2, 97.5	D 978 收发	
	L12	6.8nH	L_0402_1005Metric	118.2, 95.2	D 978 收发	
	L13	7.5nH	L_0402_1005Metric	110.7, 92.2	D 978 收发	
	L14	100nH	L_0402_1005Metric	101.1, 70.6	E 1090 接收链	
	L15	33nH	L_0402_1005Metric	57.6, 68.7	C 传感器 + GNSS	
	L16	1uH	L_Bourns-SRN4018	139.7, 83.6	A 电源	
	L17	4.7uH XEL4030-472MEC	L_Coilcraft_XxL4030	144.8, 95.0	A 电源	
	L18	100nH	L_0402_1005Metric	126.0, 93.9	G 航空波段 AIRBAND	
	D2	TPESD8L3.3 0.3pFtyp 0.5pFmax	D_0402_1005Metric	101.1, 72.6	E 1090 接收链	
	D3	TPESD8L3.3 0.3pFtyp 0.5pFmax	D_0402_1005Metric	117.5, 89.9	D 978 收发	
	D4	TPESD8L3.3 0.3pFtyp 0.5pFmax	D_0402_1005Metric	139.3, 75.4	B MCU + Flash	
	D5	TPESD8L3.3 0.3pFtyp 0.5pFmax	D_0402_1005Metric	139.3, 77.8	B MCU + Flash	
	D6	TPESD8L3.3 0.3pFtyp 0.5pFmax	D_0402_1005Metric	126.4, 89.3	G 航空波段 AIRBAND	有极性，认阴极标记带
	D7	1N4148WS	D_SOD-323	80.9, 87.6	A 电源	有极性，认阴极标记带
	D8	LED 红/黄(Vf2.0-2.2)	LED_0805_2012Metric	148.5, 80.3	A 电源	有极性，认阴极标记带
	D9	LED 红/黄(Vf2.0-2.2)	LED_0805_2012Metric	148.5, 84.2	A 电源	有极性，认阴极标记带
	F2	6V/200mA	Fuse_0805_2012Metric	122.0, 91.0	D 978 收发	
	F3	6V/200mA	Fuse_0805_2012Metric	98.3, 71.7	E 1090 接收链	
	F4	6V/200mA	Fuse_0805_2012Metric	60.3, 75.9	C 传感器 + GNSS	
	F5	6V/200mA	Fuse_0805_2012Metric	60.3, 69.2	C 传感器 + GNSS	
	Q2	AO3401A	SOT-23	122.0, 93.9	D 978 收发	
	Q3	AO3401A	SOT-23	95.3, 71.7	E 1090 接收链	
	Q4	AO3401A	SOT-23	63.5, 75.9	C 传感器 + GNSS	
	Q5	AO3401A	SOT-23	63.5, 69.2	C 传感器 + GNSS	
	U1	ME6211C33M5	SOT-23-5	74.7, 97.8	A 电源	认 pin1（缺角/圆点）
	U2	TPS7A2033PDBVR	SOT-23-5	82.3, 97.8	A 电源	认 pin1（缺角/圆点）
	U3	TPS7A2033PDBVR	SOT-23-5	89.2, 97.8	A 电源	认 pin1（缺角/圆点）
	U4	BNO085	BNO085_LGA-28	59.7, 84.5	C 传感器 + GNSS	认 pin1（缺角/圆点）
	U5	BMP388	BMP388_LGA-10	61.0, 88.9	C 传感器 + GNSS	认 pin1（缺角/圆点）
	U6	QMC5883P	LGA-16_3x3mm_P0.5mm	59.7, 96.5	C 传感器 + GNSS	认 pin1（缺角/圆点）
	U7	ATGM336H-6N-74	ATGM336H_LCC-18	73.7, 89.1	C 传感器 + GNSS	认 pin1（缺角/圆点）
	U8	RP2040	QFN-56-1EP_7x7mm_P0.4mm_EP3.	86.4, 76.9	B MCU + Flash	认 pin1（缺角/圆点）
	U9	W25Q128JVSIQ	SOIC-8_5.3x5.3mm_P1.27mm	93.7, 85.9	B MCU + Flash	认 pin1（缺角/圆点）
	U10	CC1312R1F3RGZR	QFN-48-1EP_7x7mm_P0.5mm_EP5.	104.5, 92.0	D 978 收发	认 pin1（缺角/圆点）

贴	位号	值 / 型号	封装	位置 (x, y)	阶段	备注
	U11	QPL9547TR7	DFN-8-1EP_2x2mm_P0.5mm_EP0.	107.7, 61.9	E 1090 接收链	认 pin1（缺角/圆点）
	U12	BGA2817	SOT-363_SC-70-6	117.1, 62.5	E 1090 接收链	认 pin1（缺角/圆点）
	U14	AD8313ARMZ	MSOP-8_3x3mm_P0.65mm	111.1, 73.4	E 1090 接收链	认 pin1（缺角/圆点）
	U15	TLV3501(TOKMAS)	SOT-23-6	100.6, 79.5	E 1090 接收链	认 pin1（缺角/圆点）
	U16	XA17-G4K	SOT-363_SC-70-6	98.0, 62.9	E 1090 接收链	认 pin1（缺角/圆点）
	U17	XA17-G4K	SOT-363_SC-70-6	68.6, 78.1	C 传感器 + GNSS	认 pin1（缺角/圆点）
	U18	CH224K	CH224K_ESSOP-10	132.5, 68.6	A 电源	认 pin1（缺角/圆点）
	U19	SY6970	QFN-24-1EP_4x4mm_P0.5mm_EP2.	134.5, 83.7	A 电源	认 pin1（缺角/圆点）
	U20	SY7069	TSOT-23-6	139.2, 95.0	A 电源	认 pin1（缺角/圆点）
	U21	BK4819QN32SC	QFN-32-1EP_4x4mm_P0.4mm_EP2.	127.5, 97.6	G 航空波段 AIRBAND	认 pin1（缺角/圆点）
	U22	PCM1808PWR	TSSOP-14_4.4x5mm_P0.65mm	122.8, 105.3	G 航空波段 AIRBAND	认 pin1（缺角/圆点）
	U23	TPS7A2033PDBVR	SOT-23-5	81.9, 93.5	A 电源	认 pin1（缺角/圆点）
	FL1	TA0970A	TA0970A_SMD3838-6	112.0, 65.5	E 1090 接收链	SAW 滤波器分输入输出，贴反不工作
	FL2	TA0970A	TA0970A_SMD3838-6	118.4, 66.5	E 1090 接收链	SAW 滤波器分输入输出，贴反不工作
	Y1	12MHz CL=10pF ABM8-272-T3	Crystal_SMD_3225-4Pin_3.2x2.5mm	85.1, 69.1	B MCU + Flash	无源晶体，转 180° 等价，长边对长边即可
	Y2	48MHz ABM8W-7pF	Crystal_SMD_3225-4Pin_3.2x2.5mm	108.3, 98.3	D 978 收发	无源晶体，转 180° 等价，长边对长边即可
	Y3	32.768kHz FC-135	Crystal_SMD_3215-2Pin_3.2x1.5mm	111.6, 87.0	D 978 收发	
	Y4	26MHz TCXO <=2.5ppm	Crystal_SMD_2520-4Pin_2.5x2.0mm	135.1, 102.5	G 航空波段 AIRBAND	有源 TCXO：pin2=GND / pin4=Vcc，转 180° 直接短路、上电即烧。顶面无标记，倒角在底面 #1 角（手册 C0.2）——贴前翻过来认，或先用一颗定出「底面倒角↔顶面打标」的对应关系再批量贴；板上丝印的实心圆点指向 pin1
	J1	J3_HAT_2x20_SMD排针	PinHeader_2x20_P2.54mm_	92.6, 106.3	F 对外接口	
	J2	U.FL_GNSS_EXT(经尾线转SMA)	U.FL_Hirose_U.FL-R-SMT-1_Vertical	54.0, 75.9	C 传感器 + GNSS	
	J4	USB-C_16P	USB_C_Receptacle_HRO_TC-31-M-12	145.8, 73.0	B MCU + Flash	
	J5	U.FL_978	U.FL_Hirose_U.FL-R-SMT-1_Vertical	118.3, 82.9	D 978 收发	
	J6	U.FL_1090_EXT(经尾线转SMA)	U.FL_Hirose_U.FL-R-SMT-1_Vertical	106.5, 67.3	E 1090 接收链	
	J7	U.FL_IFA_TEST(π后调试口)	U.FL_Hirose_U.FL-R-SMT-1_Vertical	90.0, 61.3	E 1090 接收链	
	J8	U.FL→内置patch	U.FL_Hirose_U.FL-R-SMT-1_Vertical	54.0, 69.2	C 传感器 + GNSS	
	J9	MX1.25WT-2P BAT	MX1.25WT-2P_1x02-1MP_P1.25mm_Horizontal	144.8, 99.0	A 电源	
	J10	U.FL_VHF(经尾线转SMA)	U.FL_Hirose_U.FL-R-SMT-1_Vertical	124.4, 82.9	G 航空波段 AIRBAND	
	ANT1	IFA_1090	ANT_IFA_1090MHz	79.2, 57.8	不贴（板载天线）	50.0mm外包络/48.5mm中心线；V4.0装盒实测 1082.5MHz、SWR 1.09
	ZP1	DNP 并-天线侧	C_0603_1608Metric	79.2, 63.6	不贴（DNP）	默认DNP；V4.0实测直通状态
	ZP2	DNP 并-电台侧	C_0603_1608Metric	84.2, 63.6	不贴（DNP）	默认DNP；历史HFSS 3.3pF仅作异常复调备选
	ZS1	0R 串	L_0603_1608Metric	81.7, 62.9	E 1090 接收链	默认0R直通；V4.0实测即为此状态
	SW1	RESET短接焊盘	SolderJumper-2_P1.3mm_Open_Pad1.0x1.	51.9, 77.5	不贴（短接焊盘）	
	SW2	BOOTSEL短接焊盘	SolderJumper-2_P1.3mm_Open_Pad1.0x1.	51.9, 81.0	不贴（短接焊盘）	
	SW3	TS24CA	SW_TS24CA_2x4_SideAct	147.9, 89.1	A 电源	
	TP1	TestPoint	TestPoint_Pad_1.0x1.0mm	51.9, 86.0	不贴（测试点焊盘）	
	TP2	TestPoint	TestPoint_Pad_1.0x1.0mm	51.9, 88.1	不贴（测试点焊盘）	
	TP3	TestPoint	TestPoint_Pad_1.0x1.0mm	51.9, 90.1	不贴（测试点焊盘）	
	TP4	TestPoint	TestPoint_Pad_1.0x1.0mm	51.9, 92.1	不贴（测试点焊盘）	
	TP5	TestPoint	TestPoint_Pad_1.0x1.0mm	51.9, 94.2	不贴（测试点焊盘）	
	TP6	TestPoint	TestPoint_Pad_1.0x1.0mm	51.9, 96.2	不贴（测试点焊盘）	
	TP7	TestPoint	TestPoint_Pad_1.0x1.0mm	51.9, 98.2	不贴（测试点焊盘）	
	H1	—	MountingHole_2.7mm_M2.5	54.0, 56.0	不贴（安装孔）	
	H2	—	MountingHole_2.7mm_M2.5	146.0, 56.0	不贴（安装孔）	

贴	位号	值 / 型号	封装	位置 (x, y)	阶段	备注
	H3	—	MountingHole_2.7mm_M2.5	54.0, 106.0	不贴（安装孔）	
	H4	—	MountingHole_2.7mm_M2.5	146.0, 106.0	不贴（安装孔）	
	BT1	MS621FE-FL11E(选贴)	MS621FE-FL11E	138.8, 105.1	A 电源	
	RT1	NCP18XH103F03RB 10k NTC	R_0603_1608Metric	54.8, 100.9	A 电源	

V4.8 BOM Verification Checklist (reference-designator order)

240 designators in total, 218 of them to be placed. Tick the box on the left after placing each part. Coordinate origin: PCB top-left corner, in mm. Chinese notes with fuller engineering context live in CHECKLIST-zh_CN.md.

Polarized / directional — reversed placement will not work | Not placed

✓	Designator	Value / Model	Footprint	Position (x, y)	Stage	Notes
	C1	10uF	C_0805_2012Metric	(70.8, 96.5)	A Power	
	C2	1uF	C_0603_1608Metric	(85.9, 99.1)	A Power	
	C3	1uF	C_0603_1608Metric	(70.8, 98.9)	A Power	
	C4	1uF	C_0603_1608Metric	(78.4, 99.3)	A Power	
	C5	1uF	C_0402_1005Metric	(85.9, 96.7)	A Power	
	C6	1uF	C_0603_1608Metric	(92.1, 97.9)	A Power	
	C10	100nF	C_0603_1608Metric	(61.0, 92.6)	C Sensors + GNSS	
	C11	100nF	C_0603_1608Metric	(64.6, 87.2)	C Sensors + GNSS	
	C12	100nF	C_0603_1608Metric	(56.5, 87.7)	C Sensors + GNSS	
	C13	100nF	C_0603_1608Metric	(59.1, 92.6)	C Sensors + GNSS	
	C14	100nF	C_0603_1608Metric	(64.6, 89.9)	C Sensors + GNSS	
	C15	4.7uF	C_0603_1608Metric	(59.9, 99.3)	C Sensors + GNSS	
	C16	100nF	C_0603_1608Metric	(64.6, 92.5)	C Sensors + GNSS	
	C17	100nF	C_0603_1608Metric	(71.4, 81.0)	C Sensors + GNSS	
	C18	10uF	C_0805_2012Metric	(74.4, 80.8)	C Sensors + GNSS	
	C19	15pF C0G	C_0603_1608Metric	(88.4, 69.1)	B MCU + Flash	
	C20	15pF C0G	C_0603_1608Metric	(79.8, 69.1)	B MCU + Flash	
	C21	100pF	C_0402_1005Metric	(103.6, 59.7)	E 1090 Receive Chain	
	C22	100nF	C_0603_1608Metric	(93.2, 75.8)	B MCU + Flash	
	C23	100nF	C_0603_1608Metric	(91.8, 81.5)	B MCU + Flash	
	C24	100nF	C_0603_1608Metric	(80.9, 80.2)	B MCU + Flash	
	C25	100nF	C_0603_1608Metric	(83.2, 71.6)	B MCU + Flash	
	C26	100nF	C_0603_1608Metric	(78.9, 74.6)	B MCU + Flash	
	C27	100nF	C_0603_1608Metric	(77.9, 82.5)	B MCU + Flash	
	C28	1uF	C_0603_1608Metric	(83.1, 83.3)	B MCU + Flash	
	C29	1uF	C_0603_1608Metric	(80.9, 83.3)	B MCU + Flash	
	C30	100pF	C_0402_1005Metric	(102.8, 67.5)	E 1090 Receive Chain	
	C31	100pF C0G	C_0402_1005Metric	(111.3, 62.2)	E 1090 Receive Chain	
	C32	100pF C0G	C_0402_1005Metric	(114.3, 62.2)	E 1090 Receive Chain	
	C33	100pF C0G	C_0402_1005Metric	(120.4, 63.2)	E 1090 Receive Chain	
	C34	100pF C0G	C_0402_1005Metric	(117.1, 73.4)	E 1090 Receive Chain	
	C35	100pF C0G	C_0402_1005Metric	(115.1, 71.1)	E 1090 Receive Chain	
	C36	100pF	C_0402_1005Metric	(106.0, 64.3)	E 1090 Receive Chain	
	C37	100nF	C_0402_1005Metric	(113.9, 77.0)	E 1090 Receive Chain	
	C38	100pF	C_0402_1005Metric	(111.6, 77.0)	E 1090 Receive Chain	
	C39	100pF	C_0402_1005Metric	(118.2, 92.8)	D 978 Transceiver	
	C40	3.6pF	C_0402_1005Metric	(110.7, 94.2)	D 978 Transceiver	
	C41	2.7pF	C_0402_1005Metric	(115.9, 97.5)	D 978 Transceiver	
	C42	6.2pF	C_0402_1005Metric	(115.9, 95.2)	D 978 Transceiver	
	C43	3pF	C_0402_1005Metric	(115.9, 92.8)	D 978 Transceiver	
	C44	3.6pF	C_0402_1005Metric	(112.8, 96.5)	D 978 Transceiver	
	C45	100pF	C_0402_1005Metric	(110.7, 90.2)	D 978 Transceiver	
	C46	3pF	C_0603_1608Metric	(104.9, 78.5)	E 1090 Receive Chain	
	C47	200pF	C_0603_1608Metric	(104.9, 80.4)	E 1090 Receive Chain	
	C48	1uF	C_0402_1005Metric	(107.4, 64.3)	E 1090 Receive Chain	
	C49	100nF	C_0603_1608Metric	(76.7, 75.4)	E 1090 Receive Chain	

✓	Designator	Value / Model	Footprint	Position (x, y)	Stage	Notes
	C51	1nF	C_0603_1608Metric	(77.0, 81.0)	E 1090 Receive Chain	
	C52	100nF	C_0402_1005Metric	(99.6, 82.3)	E 1090 Receive Chain	
	C53	100pF	C_0402_1005Metric	(93.6, 62.9)	E 1090 Receive Chain	See the Chinese companion for the detailed engineering note
	C54	100pF	C_0402_1005Metric	(103.9, 61.7)	E 1090 Receive Chain	
	C55	100pF	C_0603_1608Metric	(97.6, 66.5)	E 1090 Receive Chain	
	C56	100pF	C_0603_1608Metric	(95.0, 66.5)	E 1090 Receive Chain	
	C57	100pF	C_0402_1005Metric	(68.3, 81.0)	C Sensors + GNSS	
	C58	100pF	C_0402_1005Metric	(57.7, 76.5)	C Sensors + GNSS	
	C59	100pF	C_0402_1005Metric	(57.6, 69.8)	C Sensors + GNSS	
	C60	22uF	C_0805_2012Metric	(99.3, 98.0)	D 978 Transceiver	
	C61	100nF	C_0603_1608Metric	(102.7, 99.1)	D 978 Transceiver	
	C62	100nF	C_0603_1608Metric	(105.4, 98.3)	D 978 Transceiver	
	C63	100nF	C_0603_1608Metric	(107.1, 84.6)	D 978 Transceiver	
	C64	100nF	C_0603_1608Metric	(100.2, 85.8)	D 978 Transceiver	
	C65	100nF	C_0603_1608Metric	(99.0, 94.7)	D 978 Transceiver	
	C66	22uF	C_0805_2012Metric	(102.9, 97.2)	D 978 Transceiver	
	C67	1uF	C_0603_1608Metric	(103.6, 85.8)	D 978 Transceiver	
	C68	18pF	C_0603_1608Metric	(113.2, 85.0)	D 978 Transceiver	
	C69	18pF	C_0603_1608Metric	(110.1, 85.0)	D 978 Transceiver	
	C70	47nF	C_0603_1608Metric	(135.7, 80.2)	A Power	
	C71	4.7uF(带电源版)	C_0603_1608Metric	(133.4, 79.0)	A Power	
	C72	10uF	C_1206_3216Metric	(130.9, 79.2)	A Power	
	C73	10uF	C_0805_2012Metric	(141.3, 88.1)	A Power	
	C74	10uF	C_0805_2012Metric	(143.3, 84.9)	A Power	
	C75	10uF	C_0805_2012Metric	(139.1, 88.1)	A Power	
	C76	22uF	C_1206_3216Metric	(135.2, 96.8)	A Power	
	C77	22uF 16V X5R	C_1206_3216Metric	(141.2, 99.7)	A Power	
	C78	1uF	C_0603_1608Metric	(134.3, 93.8)	A Power	
	C79	1uF	C_0603_1608Metric	(130.6, 65.0)	A Power	
	C80	1uF 25V	C_0603_1608Metric	(136.3, 74.6)	A Power	
	C81	470pF C0G	C_0402_1005Metric	(115.2, 64.5)	E 1090 Receive Chain	
	C82	100nF	C_0603_1608Metric	(91.4, 67.5)	B MCU + Flash	
	C83	100nF	C_0603_1608Metric	(91.4, 69.8)	B MCU + Flash	
	C84	1uF	C_0603_1608Metric	(83.3, 85.6)	B MCU + Flash	
	C85	100nF	C_0603_1608Metric	(113.7, 80.5)	B MCU + Flash	
	C86	100nF	C_0603_1608Metric	(121.4, 88.1)	B MCU + Flash	
	C87	10nF C0G	C_0603_1608Metric	(84.0, 88.5)	B MCU + Flash	
	C88	100pF	C_0603_1608Metric	(126.4, 90.7)	G Airband	
	C89	18pF	C_0603_1608Metric	(128.7, 93.9)	G Airband	
	C91	10uF	C_0805_2012Metric	(129.0, 108.2)	G Airband	
	C92	10uF	C_0805_2012Metric	(121.3, 100.8)	G Airband	
	C93	10uF	C_0805_2012Metric	(124.2, 109.4)	G Airband	
	C94	100nF	C_0603_1608Metric	(120.5, 109.4)	G Airband	
	C95	100nF	C_0603_1608Metric	(117.2, 99.6)	G Airband	
	C96	100nF	C_0603_1608Metric	(128.0, 102.1)	G Airband	
	C97	1uF	C_0603_1608Metric	(129.8, 102.1)	G Airband	
	C98	100nF	C_0603_1608Metric	(131.6, 102.1)	G Airband	
	C99	100nF	C_0402_1005Metric	(133.6, 100.1)	G Airband	
	C100	100nF	C_0603_1608Metric	(119.5, 100.7)	G Airband	

✓	Designator	Value / Model	Footprint	Position (x, y)	Stage	Notes
	C101	1uF	C_0603_1608Metric	(81.1, 90.7)	A Power	
	C102	1uF	C_0603_1608Metric	(85.4, 95.1)	A Power	
	R1	10k	R_0603_1608Metric	(55.2, 84.5)	C Sensors + GNSS	
	R2	4.7k	R_0603_1608Metric	(64.1, 97.8)	C Sensors + GNSS	
	R3	4.7k	R_0603_1608Metric	(64.1, 95.2)	C Sensors + GNSS	
	R4	1k	R_0603_1608Metric	(82.0, 69.1)	B MCU + Flash	
	R5	10k	R_0603_1608Metric	(80.0, 73.2)	B MCU + Flash	
	R6	10k	R_0603_1608Metric	(87.9, 86.4)	B MCU + Flash	
	R7	5.1k DNP(带电源版)	R_0603_1608Metric	(141.4, 62.5)	Not placed (DNP)	
	R8	5.1k DNP(带电源版)	R_0603_1608Metric	(139.4, 62.5)	Not placed (DNP)	
	R9	27R	R_0603_1608Metric	(87.9, 83.3)	B MCU + Flash	
	R10	27R	R_0603_1608Metric	(85.3, 83.3)	B MCU + Flash	
	R11	3.32k	R_0402_1005Metric	(105.1, 59.7)	E 1090 Receive Chain	
	R17	10k	R_0603_1608Metric	(122.0, 97.4)	D 978 Transceiver	
	R18	10k	R_0603_1608Metric	(91.4, 71.4)	E 1090 Receive Chain	
	R19	52.3R	R_0402_1005Metric	(115.1, 73.4)	E 1090 Receive Chain	
	R21	0R	R_0603_1608Metric	(105.0, 72.6)	E 1090 Receive Chain	
	R22	0R	R_0603_1608Metric	(100.3, 66.5)	E 1090 Receive Chain	
	R23	0R	R_0603_1608Metric	(93.2, 77.2)	E 1090 Receive Chain	
	R24	0R DNP(旁路外接)	R_0402_1005Metric	(101.1, 63.1)	Not placed (DNP)	
	R25	0R DNP(旁路板载)	R_0402_1005Metric	(95.5, 62.9)	Not placed (DNP)	
	R26	10k	R_0603_1608Metric	(67.3, 73.7)	C Sensors + GNSS	
	R27	10k	R_0603_1608Metric	(67.3, 71.1)	C Sensors + GNSS	
	R30	1k DNP	R_0402_1005Metric	(96.8, 77.2)	Not placed (DNP)	
	R31	1k	R_0603_1608Metric	(101.6, 75.7)	E 1090 Receive Chain	
	R32	10k	R_0603_1608Metric	(104.9, 75.7)	E 1090 Receive Chain	
	R33	100k	R_0603_1608Metric	(100.9, 82.7)	E 1090 Receive Chain	
	R34	10k	R_0603_1608Metric	(77.0, 73.2)	E 1090 Receive Chain	
	R35	10k	R_0603_1608Metric	(77.0, 78.4)	E 1090 Receive Chain	
	R36	1k DNP	R_0603_1608Metric	(96.5, 81.8)	Not placed (DNP)	
	R37	0R	R_0603_1608Metric	(137.0, 66.4)	A Power	
	R38	180R	R_0603_1608Metric	(133.1, 88.2)	A Power	
	R39	5.62k	R_0603_1608Metric	(136.2, 88.2)	A Power	
	R40	31.6k	R_0603_1608Metric	(134.6, 88.2)	A Power	
	R41	470k	R_0603_1608Metric	(139.1, 98.6)	A Power	
	R42	150k	R_0603_1608Metric	(137.4, 98.6)	A Power	
	R43	1k	R_0603_1608Metric	(127.5, 65.0)	A Power	
	R44	10k	R_0603_1608Metric	(137.0, 69.5)	A Power	
	R45	10k	R_0603_1608Metric	(129.2, 84.0)	A Power	
	R46	10k	R_0603_1608Metric	(129.2, 86.5)	A Power	
	R47	10k	R_0603_1608Metric	(98.5, 91.7)	D 978 Transceiver	
	R48	0R	R_0603_1608Metric	(126.8, 67.0)	A Power	
	R49	0R	R_0603_1608Metric	(126.8, 69.4)	A Power	
	R50	30k	R_0603_1608Metric	(86.5, 92.8)	B MCU + Flash	
	R51	10k	R_0603_1608Metric	(85.5, 88.5)	B MCU + Flash	
	R52	10k	R_0402_1005Metric	(64.4, 85.0)	C Sensors + GNSS	
	R53	10k	R_0402_1005Metric	(64.3, 83.0)	C Sensors + GNSS	
	R54	10R	R_0402_1005Metric	(116.5, 77.0)	E 1090 Receive Chain	
	R55	10k	R_0603_1608Metric	(57.4, 89.8)	C Sensors + GNSS	

✓	Designator	Value / Model	Footprint	Position (x, y)	Stage	Notes
	R56	10k	R_0603_1608Metric	(98.5, 90.0)	D 978 Transceiver	
	R57	33R	R_0603_1608Metric	(104.0, 82.5)	E 1090 Receive Chain	
	R58	10k	R_0603_1608Metric	(132.5, 108.2)	G Airband	
	R59	10k	R_0603_1608Metric	(132.5, 105.6)	G Airband	
	R60	0R	R_0603_1608Metric	(124.0, 101.5)	G Airband	
	R61	10k	R_0402_1005Metric	(131.1, 97.9)	G Airband	
	R63	2.2k	R_0603_1608Metric	(78.4, 97.8)	A Power	
	R64	1k	R_0603_1608Metric	(145.8, 80.3)	A Power	
	R65	1k	R_0603_1608Metric	(145.8, 84.2)	A Power	
	L1	18nH 0402CS-18NXGRW	L_0402_1005Metric	(110.4, 59.8)	E 1090 Receive Chain	
	L2	33nH	L_0402_1005Metric	(57.7, 75.4)	C Sensors + GNSS	
	L7	6.8uH	L_0805_2012Metric	(97.3, 94.7)	D 978 Transceiver	
	L8	100nH	L_0402_1005Metric	(119.3, 90.3)	D 978 Transceiver	
	L9	7.5nH	L_0402_1005Metric	(112.3, 94.8)	D 978 Transceiver	
	L10	27nH	L_0402_1005Metric	(110.9, 95.7)	D 978 Transceiver	
	L11	6.8nH	L_0402_1005Metric	(118.2, 97.5)	D 978 Transceiver	
	L12	6.8nH	L_0402_1005Metric	(118.2, 95.2)	D 978 Transceiver	
	L13	7.5nH	L_0402_1005Metric	(110.7, 92.2)	D 978 Transceiver	
	L14	100nH	L_0402_1005Metric	(101.1, 70.6)	E 1090 Receive Chain	
	L15	33nH	L_0402_1005Metric	(57.6, 68.7)	C Sensors + GNSS	
	L16	1uH	L_Bourns-SRN4018	(139.7, 83.6)	A Power	
	L17	4.7uH XEL4030-472MEC	L_Coilcraft_XxL4030	(144.8, 95.0)	A Power	
	L18	100nH	L_0402_1005Metric	(126.0, 93.9)	G Airband	
	D2	TPESD8L3.3 0.3pFtyp 0.5pFmax	D_0402_1005Metric	(101.1, 72.6)	E 1090 Receive Chain	Polarized; follow the cathode marking
	D3	TPESD8L3.3 0.3pFtyp 0.5pFmax	D_0402_1005Metric	(117.5, 89.9)	D 978 Transceiver	Polarized; follow the cathode marking
	D4	TPESD8L3.3 0.3pFtyp 0.5pFmax	D_0402_1005Metric	(139.3, 75.4)	B MCU + Flash	Bidirectional low-capacitance USB ESD protector; no polarity
	D5	TPESD8L3.3 0.3pFtyp 0.5pFmax	D_0402_1005Metric	(139.3, 77.8)	B MCU + Flash	Bidirectional low-capacitance USB ESD protector; no polarity
	D6	TPESD8L3.3 0.3pFtyp 0.5pFmax	D_0402_1005Metric	(126.4, 89.3)	G Airband	See the Chinese companion for the detailed engineering note
	D7	1N4148WS	D_SOD-323	(80.9, 87.6)	A Power	See the Chinese companion for the detailed engineering note
	D8	LED 红/黄(Vf2.0-2.2)	LED_0805_2012Metric	(148.5, 80.3)	A Power	See the Chinese companion for the detailed engineering note
	D9	LED 红/黄(Vf2.0-2.2)	LED_0805_2012Metric	(148.5, 84.2)	A Power	See the Chinese companion for the detailed engineering note
	F2	6V/200mA	Fuse_0805_2012Metric	(122.0, 91.0)	D 978 Transceiver	
	F3	6V/200mA	Fuse_0805_2012Metric	(98.3, 71.7)	E 1090 Receive Chain	
	F4	6V/200mA	Fuse_0805_2012Metric	(60.3, 75.9)	C Sensors + GNSS	
	F5	6V/200mA	Fuse_0805_2012Metric	(60.3, 69.2)	C Sensors + GNSS	
	Q2	AO3401A	SOT-23	(122.0, 93.9)	D 978 Transceiver	
	Q3	AO3401A	SOT-23	(95.3, 71.7)	E 1090 Receive Chain	
	Q4	AO3401A	SOT-23	(63.5, 75.9)	C Sensors + GNSS	
	Q5	AO3401A	SOT-23	(63.5, 69.2)	C Sensors + GNSS	
	U1	ME6211C33M5	SOT-23-5	(74.7, 97.8)	A Power	Observe pin 1 / signal direction
	U2	TPS7A2033PDBVR	SOT-23-5	(82.3, 97.8)	A Power	Observe pin 1 / signal direction
	U3	TPS7A2033PDBVR	SOT-23-5	(89.2, 97.8)	A Power	Observe pin 1 / signal direction
	U4	BNO085	BNO085_LGA-28	(59.7, 84.5)	C Sensors + GNSS	Observe pin 1 / signal direction
	U5	BMP388	BMP388_LGA-10	(61.0, 88.9)	C Sensors + GNSS	Observe pin 1 / signal direction
	U6	QMC5883P	LGA-16_3x3mm_P0.5mm	(59.7, 96.5)	C Sensors + GNSS	Observe pin 1 / signal direction

✓	Designator	Value / Model	Footprint	Position (x, y)	Stage	Notes
	U7	ATGM336H-6N-74	ATGM336H_LCC-18	(73.7, 89.1)	C Sensors + GNSS	Observe pin 1 / signal direction
	U8	RP2040	QFN-56-1EP_7x7mm_P0.4m	(86.4, 76.9)	B MCU + Flash	Observe pin 1 / signal direction
	U9	W25Q128JVSIQ	SOIC-8_5.3x5.3mm_P1.27	(93.7, 85.9)	B MCU + Flash	Observe pin 1 / signal direction
	U10	CC1312R1F3RGZR	QFN-48-1EP_7x7mm_P0.5m	(104.5, 92.0)	D 978 Transceiver	Observe pin 1 / signal direction
	U11	QPL9547TR7	DFN-8-1EP_2x2mm_P0.5mm	(107.7, 61.9)	E 1090 Receive Chain	Observe pin 1 / signal direction
	U12	BGA2817	SOT-363_SC-70-6	(117.1, 62.5)	E 1090 Receive Chain	Observe pin 1 / signal direction
	U14	AD8313ARMZ	MSOP-8_3x3mm_P0.65mm	(111.1, 73.4)	E 1090 Receive Chain	Observe pin 1 / signal direction
	U15	TLV3501(TOKMAS)	SOT-23-6	(100.6, 79.5)	E 1090 Receive Chain	Observe pin 1 / signal direction
	U16	XA17-G4K	SOT-363_SC-70-6	(98.0, 62.9)	E 1090 Receive Chain	Observe pin 1 / signal direction
	U17	XA17-G4K	SOT-363_SC-70-6	(68.6, 78.1)	C Sensors + GNSS	Observe pin 1 / signal direction
	U18	CH224K	CH224K_ESSOP-10	(132.5, 68.6)	A Power	Observe pin 1 / signal direction
	U19	SY6970	QFN-24-1EP_4x4mm_P0.5m	(134.5, 83.7)	A Power	Observe pin 1 / signal direction
	U20	SY7069	TSOT-23-6	(139.2, 95.0)	A Power	Observe pin 1 / signal direction
	U21	BK4819QN32SC	QFN-32-1EP_4x4mm_P0.4m	(127.5, 97.6)	G Airband	Observe pin 1 / signal direction
	U22	PCM1808PWR	TSSOP-14_4.4x5mm_P0.65	(122.8, 105.3)	G Airband	Observe pin 1 / signal direction
	U23	TPS7A2033PDBVR	SOT-23-5	(81.9, 93.5)	A Power	Observe pin 1 / signal direction
	FL1	TA0970A	TA0970A_SMD3838-6	(112.0, 65.5)	E 1090 Receive Chain	See the Chinese companion for the detailed engineering note
	FL2	TA0970A	TA0970A_SMD3838-6	(118.4, 66.5)	E 1090 Receive Chain	See the Chinese companion for the detailed engineering note
	Y1	12MHz CL=10pF ABM8-272-T3	Crystal_SMD_3225-4Pin_	(85.1, 69.1)	B MCU + Flash	Passive crystal; 180-degree rotation is electrically equivalent
	Y2	48MHz ABM8W-7pF	Crystal_SMD_3225-4Pin_	(108.3, 98.3)	D 978 Transceiver	Passive crystal; 180-degree rotation is electrically equivalent
	Y3	32.768kHz FC-135	Crystal_SMD_3215-2Pin_	(111.6, 87.0)	D 978 Transceiver	
	Y4	26MHz TCXO <=2.5ppm	Crystal_SMD_2520-4Pin_	(135.1, 102.5)	G Airband	ACTIVE TCXO: pin2=GND / pin4=Vcc — rotating it 180 degrees shorts them and destroys the part on power-up. There is NO marking on the top face; the 0.2mm chamfer is on the BOTTOM at corner #1 (datasheet C0.2) — check it before placing, or use one unit to establish the bottom-chamfer / top-marking relationship first. The filled silkscreen dot on the board points at pin 1
	J1	J3_HAT_2x20_SMD排针	PinHeader_2x20_P2.54mm	(92.6, 106.3)	F External Interface	
	J2	U.FL_GNSS_EXT(经尾线转SMA)	U.FL_Hirose_U.FL-R-SMT	(54.0, 75.9)	C Sensors + GNSS	
	J4	USB-C_16P	USB_C_Receptacle_HRO_T	(145.8, 73.0)	B MCU + Flash	
	J5	U.FL_978	U.FL_Hirose_U.FL-R-SMT	(118.3, 82.9)	D 978 Transceiver	
	J6	U.FL_1090_EXT(经尾线转SMA)	U.FL_Hirose_U.FL-R-SMT	(106.5, 67.3)	E 1090 Receive Chain	
	J7	U.FL_IFA_TEST(π后调试口)	U.FL_Hirose_U.FL-R-SMT	(90.0, 61.3)	E 1090 Receive Chain	
	J8	U.FL→内置patch	U.FL_Hirose_U.FL-R-SMT	(54.0, 69.2)	C Sensors + GNSS	
	J9	MX1.25WT-2P BAT	MX1.25WT-2P_1x02-1MP_P	(144.8, 99.0)	A Power	
	J10	U.FL_VHF(经尾线转SMA)	U.FL_Hirose_U.FL-R-SMT	(124.4, 82.9)	G Airband	
	ANT1	IFA_1090	ANT_IFA_1090MHz	(79.2, 57.8)	Not placed (on-board antenna; per-cut measured 50.0mm outer envelope / 48.5mm centerline span)	Per-cut measured 50.0mm outer envelope / 48.5mm centerline span
	ZP1	DNP 并-天线侧	C_0603_1608Metric	(79.2, 63.6)	Not placed (DNP)	DNP by default; see the Chinese companion for tuning details
	ZP2	DNP 并-电台侧	C_0603_1608Metric	(84.2, 63.6)	Not placed (DNP)	DNP by default; see the Chinese companion for tuning details
	ZS1	0R 串	L_0603_1608Metric	(81.7, 62.9)	E 1090 Receive Chain	See the Chinese companion for the detailed engineering note
	SW1	RESET短接焊盘	SolderJumper-2_P1.3mm_	(51.9, 77.5)	Not placed (solder-bridge pad)	
	SW2	BOOTSEL短接焊盘	SolderJumper-2_P1.3mm_	(51.9, 81.0)	Not placed (solder-bridge pad)	
	SW3	TS24CA	SW_TS24CA_2x4_SideAct	(147.9, 89.1)	A Power	

✓	Designator	Value / Model	Footprint	Position (x, y)	Stage	Notes
	TP1	TestPoint	TestPoint_Pad_1.0x1.0m	(51.9, 86.0)	Not placed (test-point pad)	
	TP2	TestPoint	TestPoint_Pad_1.0x1.0m	(51.9, 88.1)	Not placed (test-point pad)	
	TP3	TestPoint	TestPoint_Pad_1.0x1.0m	(51.9, 90.1)	Not placed (test-point pad)	
	TP4	TestPoint	TestPoint_Pad_1.0x1.0m	(51.9, 92.1)	Not placed (test-point pad)	
	TP5	TestPoint	TestPoint_Pad_1.0x1.0m	(51.9, 94.2)	Not placed (test-point pad)	
	TP6	TestPoint	TestPoint_Pad_1.0x1.0m	(51.9, 96.2)	Not placed (test-point pad)	
	TP7	TestPoint	TestPoint_Pad_1.0x1.0m	(51.9, 98.2)	Not placed (test-point pad)	
	H1	—	MountingHole_2.7mm_M2.	(54.0, 56.0)	Not placed (mounting hole)	
	H2	—	MountingHole_2.7mm_M2.	(146.0, 56.0)	Not placed (mounting hole)	
	H3	—	MountingHole_2.7mm_M2.	(54.0, 106.0)	Not placed (mounting hole)	
	H4	—	MountingHole_2.7mm_M2.	(146.0, 106.0)	Not placed (mounting hole)	
	BT1	MS621FE-FL11E(选贴)	MS621FE-FL11E	(138.8, 105.1)	A Power	
	RT1	NCP18XH103F03RB 10k NTC	R_0603_1608Metric	(54.8, 100.9)	A Power	